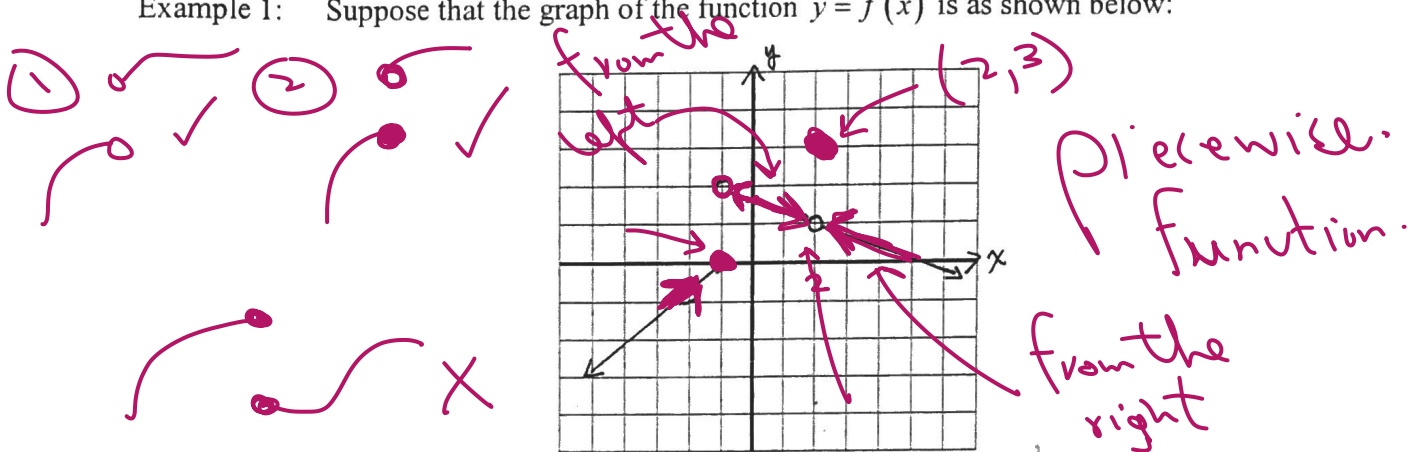


One-Sided Limits

Example 1: Suppose that the graph of the function $y = f(x)$ is as shown below:



a. Find: $f(-1) = 0$ $f(2) = 3$

b. Consider the behaviour of $f(x)$ near $x = 2$

As x approaches 2 from the left ($x < 2$) we see that $f(x)$ approaches 1.

We write this using the following notation:

$$\lim_{x \rightarrow 2^-} f(x) = 1$$

and call this the left hand limit.

As x approaches 2 from the right ($x > 2$) we see that $f(x)$ approaches 1.

We write this using the following notation:

$$\lim_{x \rightarrow 2^+} f(x) = 1$$

and call this the right hand limit.

In this case $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) \Rightarrow \lim_{x \rightarrow 2} f(x) \text{ exists} = 1$

c. Consider the behaviour of $f(x)$ near $x = -1$ and complete the following:

$$\lim_{x \rightarrow -1^-} f(x) = 0$$

$$\lim_{x \rightarrow -1^+} f(x) = 2$$

Thus, in this case $\lim_{x \rightarrow -1^-} f(x) \neq \lim_{x \rightarrow -1^+} f(x) \therefore \lim_{x \rightarrow -1} f(x) \text{ DNE}$

To test whether or not $\lim_{x \rightarrow a} f(x)$ exists, use the following criteria:

If $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$, then $\lim_{x \rightarrow a} f(x)$ does not exist. (DNE)

If $\lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x)$, then $\lim_{x \rightarrow a} f(x) = L$.

Therefore for Example 1 we may say that:

$\lim_{x \rightarrow 2} f(x) = 1$ and $\lim_{x \rightarrow -1} f(x)$ does not exist (DNE)

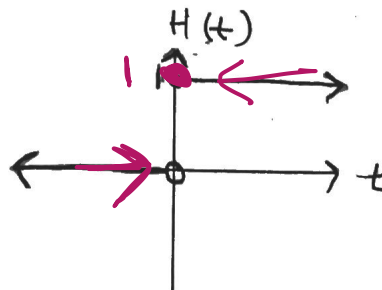
Example 2: The Heaviside function H is defined by

$$H(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

It is named after the electrical engineer Oliver Heaviside and can be used to describe an electric current that is switched on at time $t = 0$.

a. Graph the function:

$$H(0) = 1$$



b. Evaluate the following if possible:

$$\lim_{t \rightarrow 0^-} H(t) = 0$$

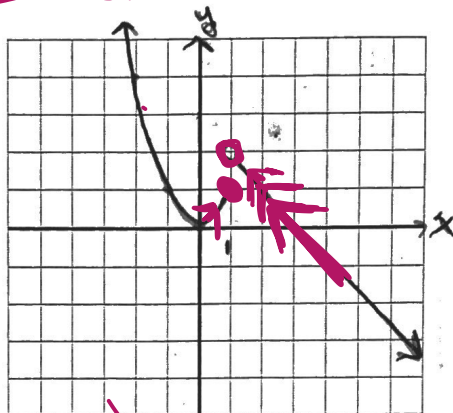
$$\lim_{t \rightarrow 0^+} H(t) = 1$$

$$\lim_{t \rightarrow 0} H(t) \text{ DNE}$$

* Since $\lim_{t \rightarrow 0^-} H(t) \neq \lim_{t \rightarrow 0^+} H(t)$ *
 $\Rightarrow \lim_{t \rightarrow 0} H(t) \text{ DNE}$

Example 3: Draw the graph of the function f defined as follows:

$$f(x) = \begin{cases} x^2, & x \in (-\infty, 1] \\ 3-x, & x \in (1, \infty) \end{cases}$$



$\lim_{x \rightarrow -2} f(x) = \text{DNE}$

$f(-2) = 3$

$\lim_{x \rightarrow 386} f(x) = -383$

Evaluate the following if possible:

$\lim_{x \rightarrow 1^+} f(x) = 2$

$\lim_{x \rightarrow 1^-} f(x) = 1$

$\lim_{x \rightarrow 1} f(x) \text{ DNE}$

$\lim_{x \rightarrow 2} f(x) = 1$

Evaluating One-Sided Limits Algebraically

Example:

If $f(x) = \begin{cases} -x-2, & x \in (-\infty, -1] \\ x, & x \in (-1, 1) \\ x^2-2x, & x \in [1, \infty) \end{cases}$



Determine whether or not $\lim_{x \rightarrow -1} f(x)$ and $\lim_{x \rightarrow 1} f(x)$ exist.

$\lim_{x \rightarrow -1^-} f(x)$

$= \lim_{x \rightarrow -1^-} -x-2$

$= -1$ ✓

$\lim_{x \rightarrow -1^+} f(x)$

$= \lim_{x \rightarrow -1^+} x$

$= -1$ ✓

$\therefore \lim_{x \rightarrow -1} f(x) = -1$

$\lim_{x \rightarrow 1^-} f(x)$

$= \lim_{x \rightarrow 1^-} x$

$= 1$ ✓

$\lim_{x \rightarrow 1^+} f(x)$

$= \lim_{x \rightarrow 1^+} x^2-2x$

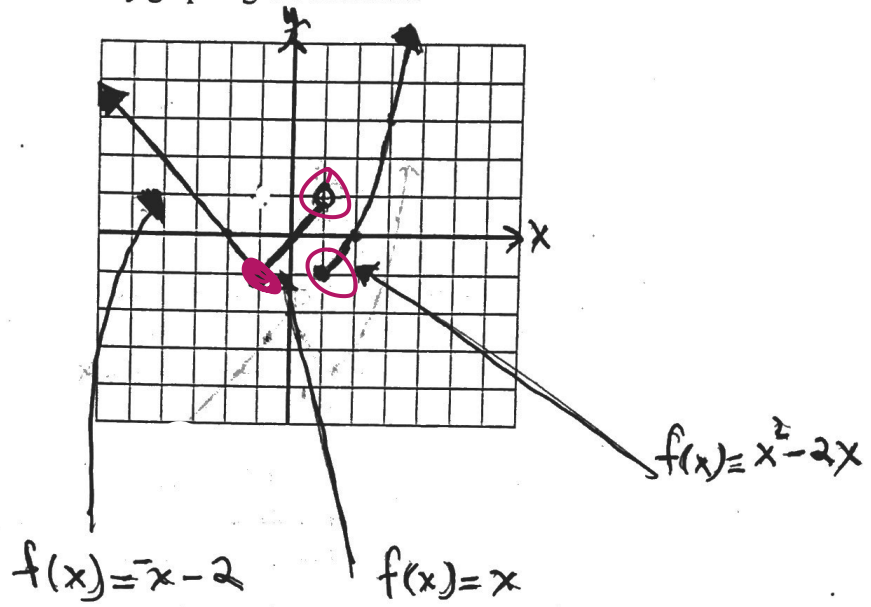
$= 1-2$

$= -1$ ✓

$\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$

$\therefore \lim_{x \rightarrow 1} f(x) \text{ DNE!}$

Verify your results by graphing the function:



NOT cont.

Domain

 $(-\infty, \infty)$

CONTINUITY

A **continuous function** is one in which there are no "gaps" or "breaks". (ie. It is smooth)

A function is continuous at a point if *the limit of the function as you approach the point equals the value of the function at this point.*

ie. A function is continuous at $x = a$ if:

- i) $f(a)$ is defined
- ii) $\lim_{x \rightarrow a} f(x)$ exists, and
- iii) $\lim_{x \rightarrow a} f(x) = f(a)$

$a \in \mathbb{R}$.

A function is continuous over an interval if it is continuous at every point in the interval.

Note: Polynomial functions are continuous at every number.

Types of Discontinuities:

* $f(a)$ is not defined

* $\lim_{x \rightarrow a} f(x)$ exists

* $f(a)$ is defined

* $\lim_{x \rightarrow a} f(x)$ exists

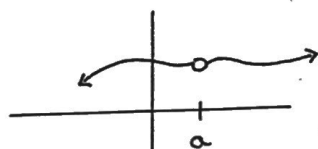
* $f(a) \neq \lim_{x \rightarrow a} f(x)$

1. Removable discontinuity
"hole"

Condition

1 or 3 will fail

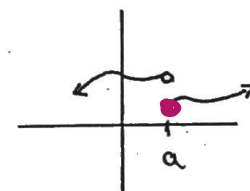
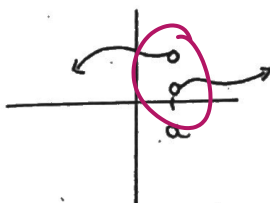
condition 2 always holds (true)



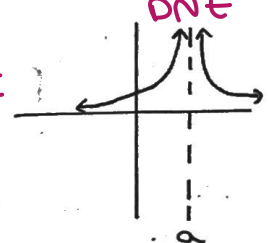
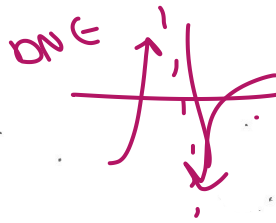
2. Jump discontinuity

$\lim_{x \rightarrow a} f(x) \text{ DNE}$

condition 2 fails.



3. Infinite discontinuity (Vertical Asymptote)

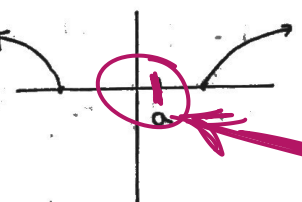
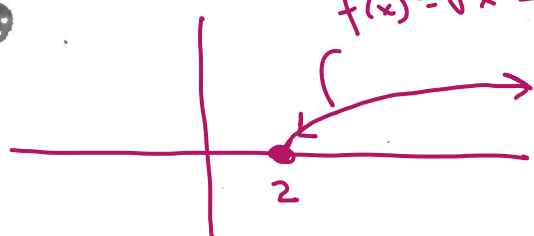


* $f(a)$ is not defined

* $\lim_{x \rightarrow a} f(x) \text{ DNE } (+\infty)$

4. Gap

$f(x) = \sqrt{x-2}$



* $f(a)$ is not defined

* $\lim_{x \rightarrow a} f(x) \text{ DNE}$

$f(2) = 0$ defined.

$$\left. \begin{aligned} \lim_{x \rightarrow 2^-} \sqrt{x-2} &= \text{DNE} \\ \lim_{x \rightarrow 2^+} \sqrt{x-2} &= 0 \end{aligned} \right\} \lim_{x \rightarrow 2} \sqrt{x-2} = \text{DNE}$$

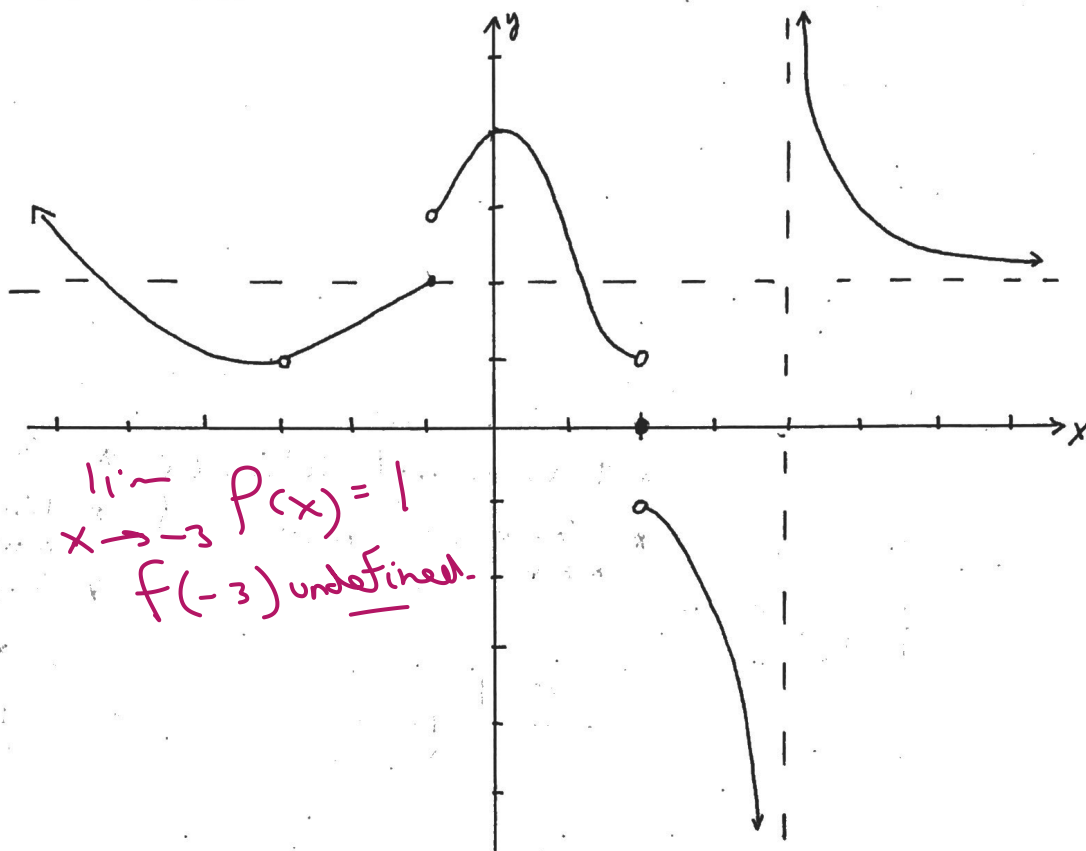
Limits

Is $f(x) = \sqrt{x-2}$ continuous

at $x=2$? NO!

$$\lim_{x \rightarrow 2} \sqrt{x-2} \text{ DNE!}$$

From the graph of $y = f(x)$ below, state the following:



$$\lim_{x \rightarrow -3} f(x) = 1$$

$$\lim_{x \rightarrow 0} f(x) = 4$$

$$\lim_{x \rightarrow -1^-} f(x) = 2$$

$$\lim_{x \rightarrow -1^+} f(x) = 3$$

$$\lim_{x \rightarrow 2^-} f(x) = 1$$

$$\lim_{x \rightarrow 2^+} f(x) = -1$$

$$\lim_{x \rightarrow 4^-} f(x) = -\infty$$

$$f(2) = 0$$

$$\lim_{x \rightarrow +\infty} f(x) = 2$$

$$\lim_{x \rightarrow -\infty} f(x) = \infty$$

Limits do not exist at:

-1, 2, 4

Discontinuities exist at:

-3, -1, 2, 4

Continuity

Example: Determine whether or not $f(x)$ is continuous on the interval $(-\infty, \infty)$.

$$f(x) = \begin{cases} (x+1)^2, & x \in (-\infty, -1] \\ x+1, & x \in (-1, 1] \\ \sqrt{x}+1, & x \in (1, \infty) \end{cases}$$

1st point @ $x = -1$

① $f(-1) = 0 \Rightarrow$ Defined ✓

② $\lim_{x \rightarrow -1^-} f(x)$

$= \lim_{x \rightarrow -1^-} (x+1)^2$
 $= 0$ ✓

$\lim_{x \rightarrow -1^+} f(x)$

$= \lim_{x \rightarrow -1^+} x+1$
 $= 0$ ✓

$\Rightarrow \lim_{x \rightarrow -1} f(x) = 0$

③ $f(-1) = \lim_{x \rightarrow -1} f(x) = 0 \Rightarrow f(x)$ is continuous at $x = -1$.

2nd point @ $x = 1$

① $f(1) = 2 \Rightarrow$ Defined

② $\lim_{x \rightarrow 1^-} f(x)$

$= \lim_{x \rightarrow 1^-} x+1$
 $= 2$ ✓

$\lim_{x \rightarrow 1^+} f(x)$

$= \lim_{x \rightarrow 1^+} \sqrt{x}+1$
 $= 2$ ✓

$\Rightarrow \lim_{x \rightarrow 1} f(x) = 2$

③ $f(1) = \lim_{x \rightarrow 1} f(x) = 2 \Rightarrow f(x)$ is continuous at $x = 1$.

$\therefore f(x)$ is continuous on $(-\infty, \infty)$